

1 Explainability in Reinforcement Learning

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Chapter 1

Latent Space Analysis

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep Learning (DL) introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, Neural Network (NN) gradually construct representations that become increasingly relevant to the task under consideration [Rumelhart et al., 1986].

These representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training. Since these representations are learned internally by the network and are not directly observable from the input or output spaces, they are commonly referred to as Latent Representation (LR). The collection of these LR learned throughout the network is referred to as a Latent Space (LS).

These LS are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of LS to encode information efficiently [Lu et al., 2019]. In Reinforcement Learning (RL), agent internal representations are used for efficient exploration in extremely open environments [Burda et al., 2018]. Recommendation systems model users and content within a shared LS to compute meaningful similarities [He et al., 2017]. Retrieval-augmented generation systems index and query knowledge bases through LR [Lewis et al., 2021]. Finally, Large Language Model (LLM) and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [Radford et al., 2021]. These examples represent only some of the most widespread applications of LR.

Given their extensive use, the study of LR has become a central topic in DL research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these LR influence model behavior.

The remainder of this chapter is organized as follows:

- Section 1.1 introduces the hypothesis that underlies much of the research on LS analysis, which we refer to as the *Concept Hypothesis* (Hyp. 1).
- Sections 1.2, 1.3, 1.4, and 1.5 examine four families of LS analysis methods, respectively based on probing, interpretable directions, feature superposition, and clustering.
- Section 1.6 concludes with a comparative synthesis of these methods and a discussion of the open questions that remain.

1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on LS analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, LS, concept, and semantics. Throughout this manuscript, we refer to this hypothesis as *Concept Hypothesis* (Hyp. 1). The goal of this section is to make this conceptual framework explicit by defining its main components.

The *Concept Hypothesis* (Hyp. 1) originates from the need to explain the generalization ability of deep NN observed after training, as discussed in Section ???. A model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y . Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [Bengio et al., 2014].

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction.

The ability of NN to perform abstraction arises from their intermediate activations. A NN does not directly map x to $\hat{y}$, it computes a sequence of intermediate activations across hidden layers. These activations are numerical encodings of the input referred to as embeddings. Once training is complete, these embeddings are structured to support the performance of a specific task. To mark this distinction, we will refer to these trained embeddings as LR, as defined in Definition 1.1.2.

Definition 1.1.2: Latent Representation

A LR is the internal encoding of an input given by an intermediate layer of a trained NN.

Each hidden layer refines the LR of the previous layer. Each transformation progressively reshapes the LR into a form more directly aligned with the output

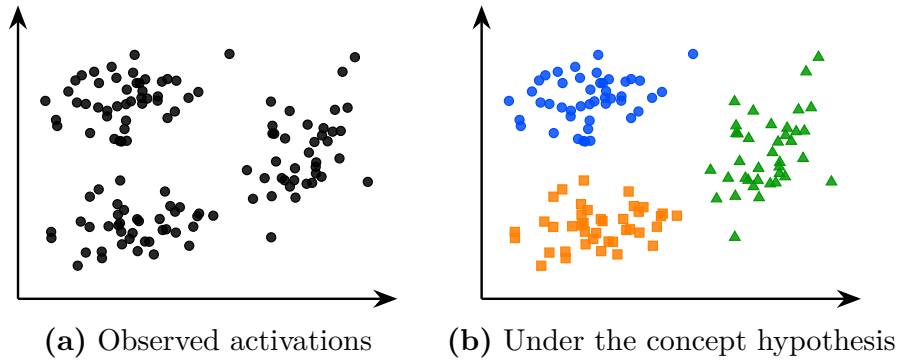


Figure 1.1: The same latent space activations seen without (a) and under (b) the *Concept Hypothesis* (Hyp. 1).

space [Rumelhart et al., 1986]. This process can be described as a sequence of transformations where each $\mathbf{h}_i$ is a LR:

$$x \rightarrow \mathbf{h}_1 \rightarrow \mathbf{h}_2 \rightarrow \cdots \rightarrow \mathbf{h}_{L-1} \rightarrow \hat{y}$$

The set of all such LR produced by a layer over a dataset defines its LS, as defined in Definition 1.1.3.

Definition 1.1.3: Latent Space

The LS is the set of all LR produced by a layer over a dataset. It is shaped by learning and organizes LR according to task-relevant properties.

Under the *Concept Hypothesis* (Hyp. 1), the analysis of the LS is assumed to provide access to abstract structures learned by the model, as illustrated in Figure 1.1. The same set of LR can be viewed either as an unstructured point cloud or, under this hypothesis, as a collection of points organized by concept membership, in which case the structure of the LS becomes meaningful and amenable to analysis. In Figure 1.1, the colors represent hypothetical concept memberships assigned by the model. In this manuscript, a concept is defined as an abstract representation that organizes information and supports reasoning (see Definition 1.1.4).

Definition 1.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

Our definition of concepts is consistent with both the pragmatic tradition in philosophy and the information-processing view of cognition. Within these frameworks, concepts arise from the need to simplify an otherwise intractable reality. Because the environment contains more information than any cognitive system can process, intelligent behavior relies on abstractions, called concepts, that discard irrelevant details while preserving the information required to achieve a given objective [Peirce, 1878, Misak, 2013]. These concepts are realized as

representations that reduce complexity and enable reasoning under limited computational resources [Newell, 1980, Simon, 1996]. Although these theories were originally developed to explain human cognition, the *Concept Hypothesis* (Hyp. 1) assumes that the same principles can be extended to deep NN.

In order to influence the model’s computations, concepts must necessarily admit a concrete realization within the LS. We refer to this realization as the Concept Structure (CS). The definition of the CS is provided in Definition 1.1.5.

Definition 1.1.5: Concept Structure

A CS is the specific form for how a concept is instantiated within the LS.

The methods reviewed in this chapter all build on the *Concept Hypothesis* (Hyp. 1) formalized in Hypothesis 1, but differ in how they characterize the CS of a concept. This choice directly determines how concepts can be identified and extracted from the LS. As a result, several families of LS analysis methods have emerged in the literature, each adopting a different view of how concepts are structured within the LS.

Hypothesis 1: Concept Hypothesis

NN generalize by performing abstraction. Through training, their LS becomes organized around concepts. Concepts are task-dependent abstractions that capture regularities relevant to the task. Analyzing the LS therefore provides access to the CS through which these concepts are instantiated in the model.

Now that the definition of a *Concept Hypothesis* (Hyp. 1) has been established, we can consider its relationship to human. According to our definition, we can imagine concepts that are useful for the machine but do not make sense to humans. This relationship is captured by the notion of semantics [Marconato et al., 2023], defined in Definition 1.1.6. Some concepts developed in LS can be readily associated with human concepts and are said to have high semantic value. Others may contribute to task performance while remaining difficult for humans to understand, and are said to have low semantic value.

Definition 1.1.6: Semantics

Semantics refers to the alignment between concepts in the model LS and concepts used by a human observer. A concept is semantic if it can be associated with a human interpretable concept.

It is important to note that this conceptual framework is not supported by a complete theoretical understanding of deep NN. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable CS within LS. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and LS analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing LS and extracting the concepts they encode. We begin with probing, one of the most direct approaches for assessing whether concepts are represented in the LS.

1.2 Probing

Probing methods aim to determine whether a given semantic concept is encoded within the LS of a NN [Li et al., 2024, Karvonen, 2024]. The idea behind these methods is that if a concept can be reliably decoded from LR, it is likely relevant to the problem being solved. Studying such concepts can therefore provide insight into how the network internally represents the problem.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example x . The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \dots, c_n^{(i)})\}_{i=1}^N.$$

The data are subsequently projected into the LS of the pretrained model. For an input x , we write $\mathbf{h}_l(x) \in \mathbb{R}^{n_l}$, or simply $\mathbf{h}_l$, the LR obtained at layer l . A classifier, referred to as a probe, is then trained to predict the presence of a given concept from this LR. It is denoted by $p_\theta^{l,j}$, where θ represents its parameters, l indicates the layer from which the LR is extracted, and j identifies the concept being predicted. The probe maps the LR to a scalar value in $[0, 1]$, which can be interpreted as the probability that the concept is present. The parameters θ are learned by minimizing a loss function over the training examples using Stochastic Gradient Descent (SGD). This gives the following mathematical formulation:

$$p_\theta^{l,j} : \mathbb{R}^{n_l} \rightarrow [0, 1], \quad \min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L} \left(p_\theta^{l,j} \left(\mathbf{h}_l(x^{(i)}) \right), c_j^{(i)} \right). \quad (1.1)$$

The architecture of the probe must be kept intentionally simple. A probe with excessive expressive capacity may be able to predict the concept by exploiting complex patterns within the representation. These patterns do not necessarily indicate that the concept is explicitly encoded in the LS. Instead, they may result from the ability of the probe itself to combine and transform the available information. Consequently, a highly expressive probe makes it difficult to distinguish between information that is directly accessible in the representation and information that is reconstructed by the classifier. Restricting the capacity of the probe therefore helps ensure that its predictions primarily reflect the information already present in the LS.

For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the NN and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the LS contains exploitable information related to the considered concept.

Figure 1.2 illustrates the decision boundaries of such linear probes for three concepts, where each colored region indicates the area classified positively by the corresponding probe.

An application of this technique can be found in studies investigating whether LLM develop internal representations of their environment [Li et al., 2024, Karvonen, 2024]. In these works, researchers train LLM architecture to play games such as chess or Othello in a purely textual setting. The model is provided only with sequences of

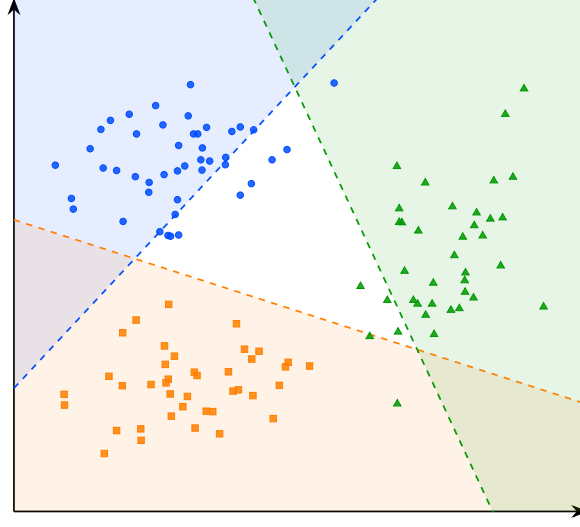


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C.

169 moves and information about the actions it can take, all expressed in textual form.
 170 The central question is whether the LLM develops an internal representation of
 171 the game state. To address this question, the probed concepts correspond to the
 172 presence or absence of different game pieces on specific board positions.

173 As illustrated in Figure 1.3, probes are able to reconstruct the full board
 174 state from the model’s LR, even though the only available information to the
 175 model is a sequence of text. In this figure, each row corresponds to a piece type,
 176 comparing the ground truth with the probe’s predictions and confidence. The
 177 close agreement between the ground truth and predicted positions suggests that
 178 the model develops a structured representation of the underlying environment
 179 from which the data are generated.

180 However, the fact that a concept can be decoded from LR does not necessarily
 181 imply that the model actually uses this information to perform its task. To assess
 182 whether LR play a causal role in the model’s output, it is possible to perform
 183 interventions on them. An intervention consists of modifying a LR in a controlled
 184 manner by activating or deactivating a given concept, and then studying its
 185 impact on the model’s output. If the resulting change in the model’s output is
 186 consistent with the applied perturbation, this suggests that the representation
 187 had a causal role.

188 To this end, the LR is minimally modified such that the probe prediction for
 189 the target concept c_j changes state, while the predictions associated with other
 190 concepts remain unchanged. This goal can be formulated as an optimization
 191 problem over the LR itself. The optimization is performed directly on the LR,
 192 using SGD. The solution to this problem is the optimal modified LR, defined as:

$$\mathbf{h}_l^* = \arg \min_{\mathbf{h}_l'} \underbrace{\gamma \mathcal{L}_{\text{flip}}(p_{\theta}^{l,j}(\mathbf{h}_l'), c_j)}_{\text{flip target concept } c_j} + \underbrace{\lambda \sum_{k \neq j} \mathcal{L}_{\text{preserve}}(p_{\theta}^{l,k}(\mathbf{h}_l'), c_k)}_{\text{preserve other concepts } c_{k \neq j}} + \underbrace{\beta \|\mathbf{h}_l' - \mathbf{h}_l\|_2^2}_{\text{minimality}}$$

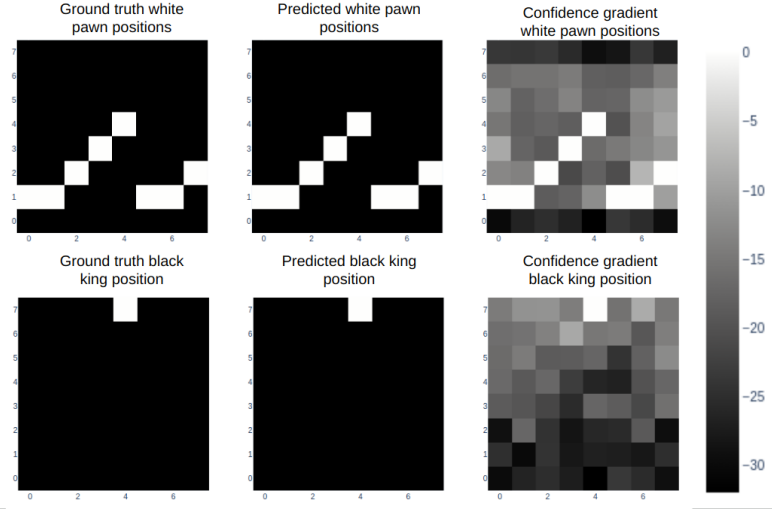


Figure 1.3: Probing analysis of a large language model trained to play chess [Karvonen, 2024].

193 where:

- 194 • $\mathbf{h}_l$ is the original LR of the input at layer l ,
- 195 • $\mathbf{h}'_l$ is the modified LR being optimized,
- 196 • $\mathbf{h}^*_l$ is the optimal modified LR,
- 197 • $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change
- 198 state (i.e., flipping the predicted label),
- 199 • $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- 200 • $\|\mathbf{h}'_l - \mathbf{h}_l\|_2^2$: regularization term ensuring the modified LR remains close to
- 201 the original one,
- 202 • γ, λ, β : weighting coefficients controlling the trade-off between the inter-
- 203 vention strength, the preservation of other concepts, and the minimality of
- 204 the perturbation.

205 In the context of LLM studied previously, such interventions have been used
 206 to remove specific pieces from the model’s internal representation of the game
 207 state [Li et al., 2024, Karvonen, 2024]. The main observation is that, in the vast
 208 majority of cases, the model behaves as if the piece had effectively disappeared.
 209 The LLM continues to generate legal moves while correctly accounting for the
 210 absence of the removed piece. This provides evidence for the causal role of these
 211 internal representations in the model’s behavior.

212 A main limitation of the probing approach is that the concepts under in-
 213 vestigation must be predefined before any analysis can be performed. This
 214 introduces a strong bias in what can be studied. Moreover, annotating a dataset
 215 for each concept is extremely time-consuming and requires significant human
 216 effort. Interventions in LS are also difficult to interpret, as it is not always clear

what constitutes a meaningful change in the network’s output under a given perturbation.

Moreover, probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a LR. This binary perspective can be overly restrictive, as concepts may be represented in a more continuous manner. To obtain a more detailed characterization of these representations, the next section turns to methods that describe a concept as a direction in LS along which it varies continuously.

1.3 Directions

Probing describes a concept as a binary attribute, either present or absent in a LR. The methods in this section instead associate each concept with a direction in the LS. Moving a LR along this direction is expected to strengthen or weaken the concept smoothly. This provides information that a binary probe cannot express.

Empirically, it has been observed that shifting a LR along such a direction can induce coherent and semantically meaningful changes in the model’s output [Haas et al., 2024, Voynov and Babenko, 2020]. Such transformations can be mathematically express as:

$$\mathbf{h}' = \mathbf{h} + \alpha \mathbf{v}_i,$$

where:

- $\mathbf{h}$ denotes the original LR,
- $\mathbf{v}_i$ is a unit vector associated with concept i ,
- α controls the strength of the intervention,
- $\mathbf{h}'$ is the resulting perturbed LR.

In many cases, the resulting change in the model’s output varies smoothly with α , suggesting an approximately linear CS with respect to certain semantic factors. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the LS.

Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be associated with specific semantic attributes. The objective of this line of work is therefore to identify directions in LS that align with meaningful variations in the encoded concepts. This is illustrated schematically in Figure 1.4, where arrows represent semantic shifts from one concept toward others, and moving along these directions induces a coherent change in the model’s behavior, suggesting that the transition between concepts is continuously encoded along these axes.

This phenomenon has been extensively studied in generative image models. As illustrated in Figure 1.5, where each row corresponds to a distinct direction $\mathbf{v}_i$ and each column to an increasing value of α , controlled perturbations of LR along interpretable directions can induce specific semantic transformations, such as stroke thickness modification in digit images, head rotation in facial images, or texture and background changes in natural images [Voynov and Babenko, 2020]. These observations have led to the development of the field of LS editing, whose

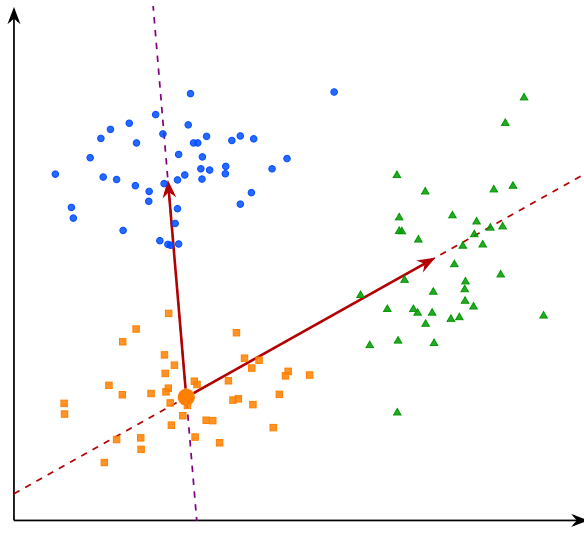


Figure 1.4: Interpretable directions in the latent space.

objective is to manipulate generated content directly through modifications in LS rather than through changes in the input prompt.

A central challenge remains identifying such interpretable directions. Most existing approaches rely on auxiliary models that encode human defined semantics, such as CLIP [Haas et al., 2024], to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in LS without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in LS, while a second model attempts to recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent CS [Voynov and Babenko, 2020]. In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations.

Despite these advances, several challenges remain. It appears that not all concepts encoded by NN are linearly identifiable in LS. In practice, interpretable direction methods typically discover only a few dozen directions per dataset, which intuitively seems insufficient to account for the full diversity of semantic variations that a generative model is capable of producing.

In addition, LR often exhibit a phenomenon known as entanglement, illustrated in Figure 1.6, where multiple semantic attributes are mixed within the same directions of the LS. For instance, in a generative face model, moving along the “glasse” direction may simultaneously modify age and gender, whereas a disentangled LR would let each axis control a single semantic attribute while leaving the others unchanged. This lack of separability suggests that semantic information is not cleanly organized in the LS. This observation motivates the *Superposition Hypothesis* (Hyp. 2), which proposes a different view of how NN



Figure 1.5: Interpretable directions in the latent space of a generative model [Voynov and Babenko, 2020].

organize information internally. The next section introduces this hypothesis.

1.4 Superposition

The previous section highlighted a limitation of interpretable directions. Only a limited number of concepts can be represented by a linear direction, and these directions are often entangled with multiple semantic concepts. To account for this, an extension of the *Concept Hypothesis* (Hyp. 1) has been proposed under the name *Superposition Hypothesis* (Hyp. 2), formalized in Hypothesis 2 [Elhage et al., 2022].

Hypothesis 2: Superposition Hypothesis

Concepts are preferentially represented by distinct linear directions in LS. When the number of concepts exceeds the available dimensions, multiple concepts are encoded in superposition within the same directions.

According to this hypothesis, in an unconstrained LS, each concept is represented in its own independent linear direction. The idealized non-superposed

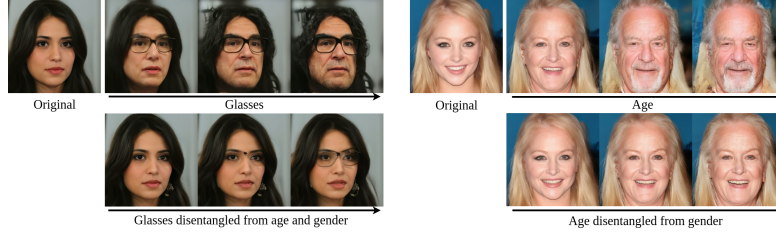


Figure 1.6: Illustration of entanglement in the latent space of a generative face model [Haas et al., 2024].

LR can be mathematically expressed as follows:

$$\mathbf{h} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \quad \|\boldsymbol{\alpha}\|_0 \ll n,$$

where:

- $\mathbf{h}$ denotes a LR,
- n is the total number of concepts,
- $\mathbf{v}_i$ represents the direction associated with concept c_i ,
- α_i corresponds to the activation strength of concept c_i in the LR $\mathbf{h}$,
- $\mathbf{v}_i^\top \mathbf{v}_j \approx 0$ indicates that concept directions are approximately orthogonal, meaning each concept independently controls a single factor of variation,
- $\|\boldsymbol{\alpha}\|_0 \ll n$ indicates that only a small fraction of concepts are active for a given LR.

However, NN operate in finite-dimensional spaces, so when the number of concepts exceeds the available dimensions, such a one-to-one correspondence becomes impractical. The network must therefore reuse its representational dimensions, allowing multiple concepts to be encoded using the same directions. This shared encoding is referred to as superposition [Bricken et al., 2023]. This phenomenon becomes an obstacle for any method that attempts to analyze LR directly in the original activation space.

Building on this observation, several works based on the *Superposition Hypothesis* (Hyp. 2) have developed methods that aim to recover less superposed LR. A common approach is to map LS into a higher-dimensional space while imposing additional constraints that encourage individual concepts to be represented separately. This transformation is illustrated in Figure 1.7, while LR are entangled in the initial LS, the sparse representation aligns each concept with a dedicated feature axis.

SAE provide a practical implementation of this approach and can be defined as follows:

$$z = f_{\text{enc}}(\mathbf{h}), \quad \hat{\mathbf{h}} = f_{\text{dec}}(z)$$

$$\mathcal{L}_{\text{auto}} = \gamma \|\mathbf{h} - \hat{\mathbf{h}}\|_2^2 + \lambda \|z\|_1$$

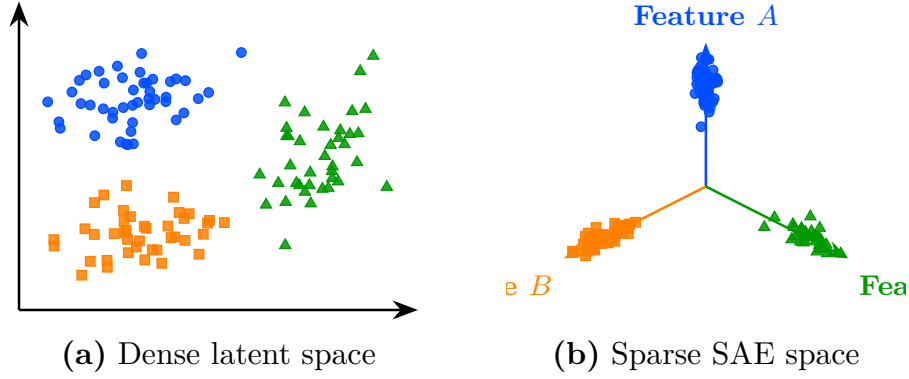


Figure 1.7: Transformation from a dense latent space (a) to a sparse Sparse Autoencoder (SAE) representation (b).

where:

- $f_{\text{enc}}(\mathbf{h})$ is the encoding function that projects the original LR $\mathbf{h}$ into a higher-dimensional sparse space,
- $f_{\text{dec}}(z)$ is the decoding function that reconstructs the original LR from the sparse code,
- $z \in \mathcal{Z}$ is the sparse representation of $\mathbf{h}$, with $\dim(z) \gg \dim(\mathbf{h})$,
- $\hat{\mathbf{h}}$ is the reconstructed LR,
- $\|\mathbf{h} - \hat{\mathbf{h}}\|_2^2$ is the reconstruction loss ensuring that the original LR can be faithfully recovered,
- $\|z\|_1$ is the sparsity regularization term that encourages most components of z to be zero,
- γ, λ are weighting coefficients controlling the trade-off between reconstruction fidelity and sparsity.

The parameters of these projection functions f_{enc} and f_{dec} are learned by minimizing $\mathcal{L}_{\text{auto}}$ using SGD. Under this formulation, each dimension of $\mathcal{Z}$ is interpreted as a dedicated concept direction.

In practice, both f_{enc} and f_{dec} are implemented as linear projections between the original LS and the high-dimensional space $\mathcal{Z}$. This choice is motivated by the same principle underlying probing methods. The transformation applied to the LR should remain as simple as possible. Using more expressive projection functions could introduce additional structures into the transformed LR. In this case, it would become difficult to determine whether an observed concept was already encoded in the model’s LR, or whether it was introduced by the projection itself. Restricting f_{enc} and f_{dec} to linear projections therefore helps ensure that the concepts identified in $\mathcal{Z}$ can be attributed to the original LR, rather than to the transformation used to analyze it.

The method proves to be powerful when successfully applied to language models [Templeton et al., 2024]. In these applications, SAE are used to desuperpose the LR of LLM. Once this LR is obtained, it has been shown that many



Figure 1.8: A concept direction corresponding to the Golden Gate Bridge, discovered by a sparse autoencoder applied to a large language model [Templeton et al., 2024].

concept vectors exhibit semantic properties. A example is the Golden Gate Bridge concept direction, illustrated in Figure 1.8, which activates systematically whenever the input text or image refers to the Golden Gate Bridge, regardless of the language or modality. This demonstrates that the learned concept is not tied to a specific surface form but captures a deeper semantic abstraction shared across modalities.

SAE also provide a way to intervene directly on the LR learned by the model. A concept can be artificially activated by modifying its corresponding feature in the sparse representation z , even when the input contains no reference to it. The modified LR is then projected back into the original activation space, after which the model proceeds with standard next-token generation from $\mathbf{h}^*$:

$$\mathbf{h}^* = f_{\text{dec}}(z^*), \quad \text{where} \quad z^* = z + \delta e_i,$$

where:

- z is the original sparse representation of the input,
- z^* is the modified sparse representation,
- e_i is the basis vector of $\mathcal{Z}$ associated with the target concept direction,
- δ controls the strength of the artificial activation,
- $\mathbf{h}^*$ is the resulting edited LR in the original activation space.

With this method, a reference to the Golden Gate Bridge can be injected into the LLM’s internal representation. This causes the model’s output to shift toward discussing the Golden Gate Bridge, even when the prompt contains no reference to it. This provides evidence that the learned feature plays a causal role in the model’s behavior. Such interventions enable control over specific model behaviors through LS editing.

Despite their empirical success, *Superposition Hypothesis* (Hyp. 2) exhibit several important limitations that question their assumptions. These limitations can be summarized as follows:

- **Instability of the decomposition.** Repeating the SAE training process multiple times on the same model does not yield the same decomposition of the LS. Different runs produce different concept directions in $\mathcal{Z}$, even when achieving comparable reconstruction performance, which raises questions about the reliability of the discovered CS.
- **Lack of semantic coverage.** In the high-dimensional space $\mathcal{Z}$, a large fraction of dimensions do not correspond to any interpretable semantic concept. This abundance of non-semantic directions raises questions about whether the decomposition truly reflects the model’s CS.
- **Sparsity–semantics trade-off.** Increasing the sparsity coefficient λ beyond a certain threshold degrades the semantic quality of the learned features [Jermyn et al., 2024]. Suggesting that enforcing excessive sparsity comes at the cost of interpretability.
- **Failure of scaling to remove superposition.** Under the *Superposition Hypothesis* (Hyp. 2), increasing the dimensionality of the LS should reduce superposition. However, this is not observed in practice. Even in very high-dimensional LS, representations remain superposed, indicating that scaling alone does not resolve the entanglement.

These limitations suggest that *Superposition Hypothesis* (Hyp. 2), while powerful, do not fully resolve the problem of CS. Convincing empirical results do not necessarily validate the underlying theoretical assumptions.

So far, every method described, whether based on probing, interpretable directions, or superposition, relies on a common assumption. The CS can be recovered from LR through a linear decomposition. This assumption has enabled substantial progress in understanding neural representations, but it also imposes a strong constraint on how concepts are expected to be represented. The following section examines clustering as an alternative approach that does not rely on this assumption.

1.5 Clustering

As the previous section noted, every LS analysis method presented so far rests on a linear view of the CS. For probing methods, concepts must be at least partially linearly separable for simple probes to detect them. Interpretable direction methods posit that variations associated with a concept can be captured by moving along a linear axis of the LS. The *Superposition Hypothesis* (Hyp. 2) proposes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to empirical successes, several observations suggest that the linearity hypothesis may not fully capture the CS.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches,

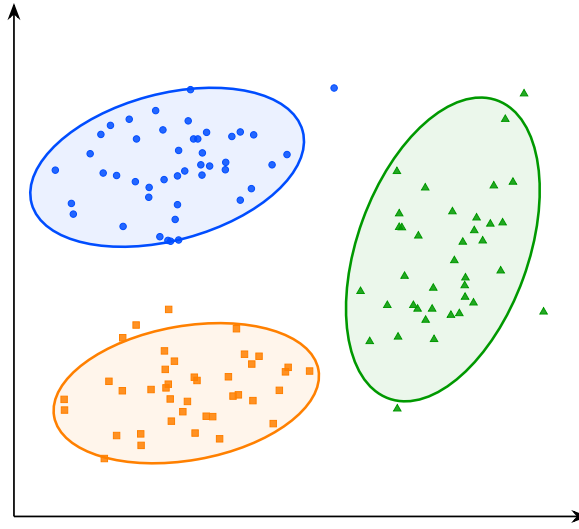


Figure 1.9: Visualization of the latent space structure.

enforcing excessive sparsity degrades interpretability. This suggests that the underlying representation may not be perfectly described by a sparse linear decomposition. Taken together, these observations indicate that semantic concepts may not always exhibit a linear CS. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of LS.

To the best of our knowledge, clustering-based analysis is the only family of LS analysis methods that does not rely on any assumption of a linear CS [Ren et al., , Wei et al.,]. Rather than assuming that concepts correspond to directions, clustering methods only assume that conceptual similar inputs produce nearby LR. Conversely, inputs that differ significantly from a concept perspective are expected to be located farther apart in the LS. This is a considerably weaker assumption, in the sense that it is empirically verified in a wide range of settings.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of LR that occupy the same region of the LS. This is shown schematically in Figure 1.9, where the points represent LR associated with concepts, and the ellipses indicate the approximate extent of each group and highlight their geometric separation.

This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is projected into a LS using a NN. The resulting LR form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct concepts.

Deep clustering has been particularly successful in unsupervised image classification. For this type of task, a specific NN architecture called an autoencoder is commonly used. An autoencoder is composed of an encoder f_{enc} and a decoder f_{dec} . The encoder maps an input $x \in \mathcal{X}$ to a low-dimensional LR $\mathbf{z} \in \mathbb{R}^d$, the LR produced by the encoder. The decoder then maps this LR back to the input space to produce a reconstruction $\hat{x}$. The model is trained by SGD to minimize the reconstruction error between the original input and its reconstruction using a Mean Squared Error (MSE) loss. This objective encourages the LS to

453 preserve the most informative features of the input while enforcing a compact
 454 representation. This can be expressed mathematically as follows:

$$x \xrightarrow{f_{\text{enc}}} \mathbf{z} \xrightarrow{f_{\text{dec}}} \hat{x}, \quad d \ll \dim(\mathcal{X}), \quad \mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2.$$

Once the autoencoder has been trained sufficiently well, the reconstruction loss $\mathcal{L}_{\text{rec}}$ should be low enough for the learned representation to be useful. A clustering algorithm can then be applied to the compressed representations of the images. Let $\mathcal{D}$ denote a dataset of inputs, through the encoder f_{enc} , this dataset is mapped to a point cloud in the LS:

$$\mathcal{Z} = \{\mathbf{z}^{(i)} \in \mathbb{R}^d \mid \mathbf{z}^{(i)} = f_{\text{enc}}(x^{(i)}), x^{(i)} \in \mathcal{D}\}.$$

455 A classical clustering algorithm is applied to the set $\mathcal{Z}$ of LR to identify
 456 groups of similar LR. Each resulting cluster can therefore be interpreted as a
 457 group of images that the model considers similar.

458 An example of this methodology can be observed in Figure 1.10, where
 459 the approach is applied to the STL-10 dataset [Coates et al., 2011] using k -
 460 means clustering [MacQueen, 1967] with a number of clusters arbitrarily fixed
 461 to $K = 10$ [Xie et al.,]. For each cluster, the corresponding line shows the 10
 462 samples closest to the associated centroid in the LS.

463 We observe that the method captures coherent semantic categories such
 464 as animals, vehicles, and flying entities. This result is particularly striking
 465 given that the only available information to the system consists of raw images.
 466 It suggests that the compression process induces a LS in which semantically
 467 meaningful CS emerge.

468 More advanced approaches combine representation learning and clustering
 469 within a single optimization process. In this setting, additional loss terms
 470 encourage LR to organize themselves around cluster centroids during training.
 471 This often produces LS that are easier to analyze and improves the quality of
 472 the discovered semantic CS [Guo et al., , Miklautz et al.,].

473 Despite its effectiveness, clustering-based analysis presents several important
 474 limitations.

475 First, as with most unsupervised semantic discovery techniques, interpreting
 476 the resulting clusters remains partly subjective. Benchmark datasets often
 477 provide class labels for evaluation, but the discovered clusters may not match
 478 the categories that humans consider relevant.

479 Second, clustering methods are highly sensitive to design choices, as the
 480 number of clusters, the distance metric, and the clustering algorithm itself can
 481 all significantly influence the results, making parameter selection challenging
 482 and often requiring substantial empirical tuning.

483 Finally, to the best of our knowledge, no intervention technique has yet been
 484 proposed in the context of clustering-based analysis. Unlike probing or SAE
 485 analysis, which allow targeted modifications of LR to study their causal role,
 486 clustering methods currently offer no mechanism to alter the model’s behavior
 487 by acting directly on the discovered CS.

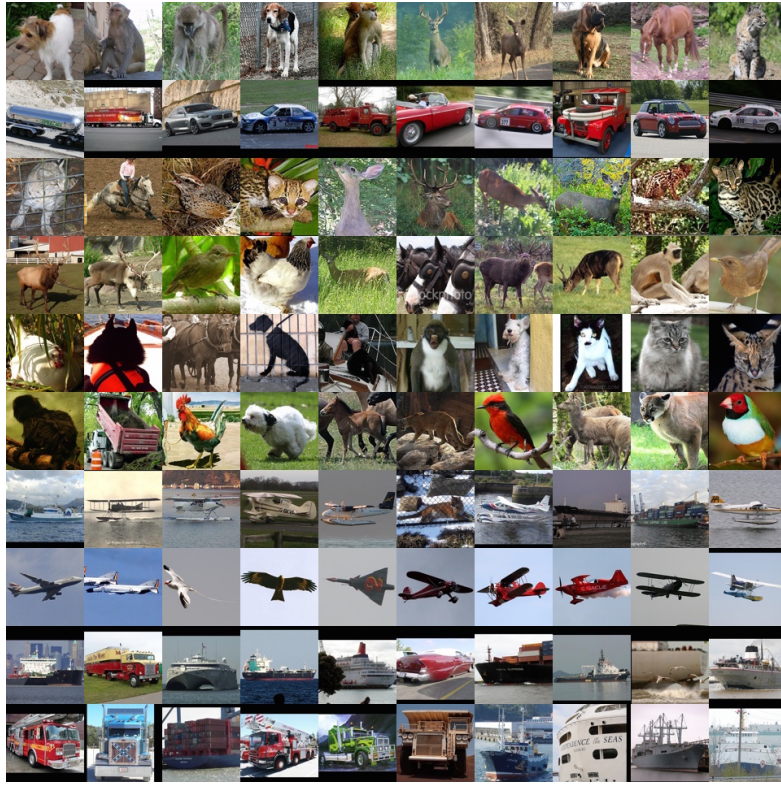


Figure 1.10: Clusters discovered by a deep clustering model applied to natural images [Xie et al.,].

1.6 Conclusion

The four families of methods reviewed in this chapter are summarized in Figure 1.11. They include probing, interpretable directions, sparse autoencoders, and clustering, all of which build upon the *Concept Hypothesis* (Hyp. 1) introduced in Section 1.1. Each family approaches the analysis of LR from a different angle. However, they share a common goal of understanding how LR are structured and what information they contain.

However, a fundamental difficulty remains. There is currently no consensus on how to rigorously evaluate whether these methods truly capture the internal concepts of a model. Most existing metrics assess reconstruction quality or linear separability, but these do not directly measure whether the discovered CS correspond to meaningful abstractions from the model’s perspective. This leaves open the question of what it means, formally, for a concept to be represented in a NN.

Addressing this question may require contributions from disciplines beyond computer science and mathematics. As the notion of semantics is inherently tied to human interpretation, a purely formal or computational account may be insufficient. Insights from philosophy, cognitive science, or psychology could prove valuable in grounding the notion of concept in a way that is both theoretically rigorous and empirically tractable. Despite these open questions, the methods

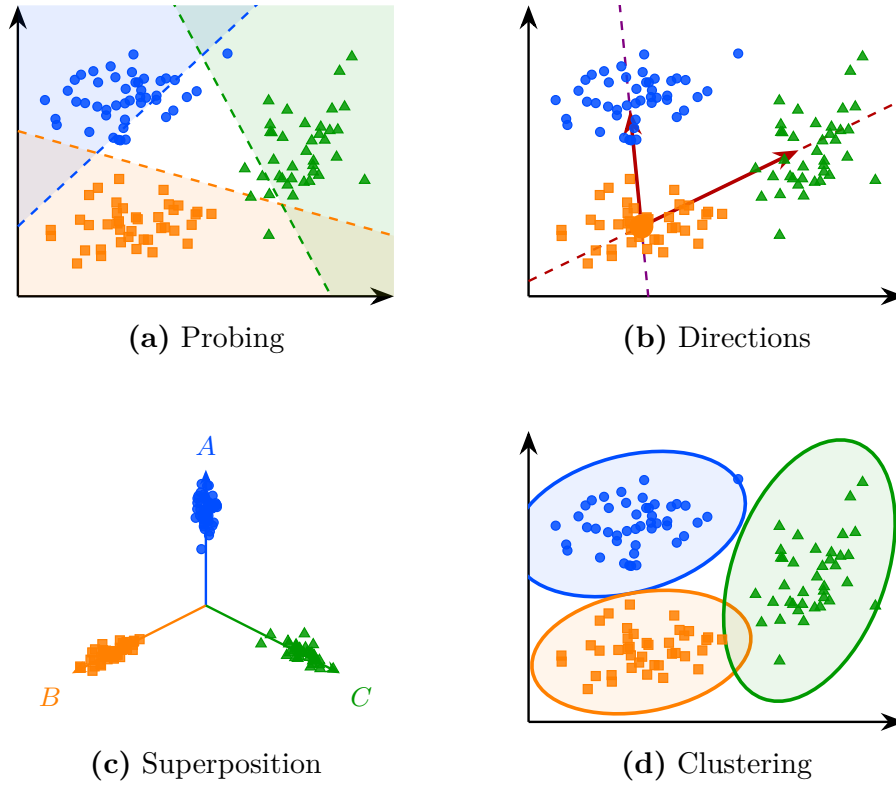


Figure 1.11: Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.

studied in this chapter provide a practical foundation for analyzing the internal representations of NN.

Building on the *Concept Hypothesis* (Hyp. 1) and the insights developed throughout this chapter, this manuscript investigates a representation-based approach to explainability in RL. To establish the necessary foundations, the next chapter introduces the principles of RL. The subsequent chapter then presents the proposed method for analyzing these representations and studying their potential for explainability in RL (Chapter ??).

516 Acronyms

517 **CS** Concept Structure 4, 8, 9, 14–17

518 **DL** Deep Learning 1

519 **LLM** Large Language Model 1, 5–7, 12, 13

520 **LR** Latent Representation 1–3, 5–9, 11–17

521 **LS** Latent Space 1–5, 7–16

522 **MSE** Mean Squared Error 15

523 **NN** Neural Network 1, 2, 4, 5, 9, 11, 15, 17, 18

524 **RL** Reinforcement Learning 1, 18

525 **SAE** Sparse Autoencoder 11–14, 16

526 **SGD** Stochastic Gradient Descent 5, 6, 12, 15

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